

Setting up the rust investigation

One sheet per group. These jars sit on the shelf for five weeks — you do not get to start again.

Group	
Our condition	
Date set up	

Steps

1. Collect three jars. Check each one for chips. If it is cracked, swap it.
2. Weigh your three nails TOGETHER. Write the mass below before they go in the jar.
3. Set up your condition in all three jars, exactly the same way each time.
4. Fill in and cut off the three labels at the bottom. Stick one on each jar.
5. Write the repeat sentence. Then put the jars on the shelf.

Why three jars and not one If you only had one and something went wrong with it, you would never know. Three lets you spot the odd one out.

Starting mass

	Jar 1	Jar 2	Jar 3
Mass of 3 nails (g)			

Record to the precision your scales actually give you. If the scales read whole grams, do not write decimal places — that is an error, not extra care.

The repeat sentence

Write the ONE sentence that would let another group set this up exactly the same way, without asking you anything. Name the nails, the liquid, how much, and whether the lid is on.

Check it against this: “Three 65 mm nails, half covered in tap water, lid off, on the shelf by the window.” Does yours give that much?

Our prediction

Which condition do you think will rust the most, and why?

Cut off these labels and stick one on each jar

RUST INVESTIGATION Group: _____ Condition: _____	RUST INVESTIGATION Group: _____ Condition: _____	RUST INVESTIGATION Group: _____ Condition: _____
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Jar 1 of 3 Nail mass: _____ g Date: _____	Jar 2 of 3 Nail mass: _____ g Date: _____	Jar 3 of 3 Nail mass: _____ g Date: _____
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Safety Glasses on. Nails counted out and counted back in. Jars stay on the bench and are carried with two hands, one at a time. Report any spill straight away and clean it up.